

Olympiad Test Paper — Mathematics (Class 7) — Paper 3

Chapter 8: Working with Fractions

Paper 3 — (progressively harder)

Subject	Mathematics (NCERT Class 7)
Chapter	8 — Working with Fractions
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Beware: dividing by a fraction below 1 enlarges, multiplying by it shrinks; chained 'of the remaining' fractions act on the leftover, not the whole; mixed numbers must become improper before multiplying or dividing; and 'flip the divisor, not the dividend' when dividing. Do not write on this paper — mark answers on your answer sheet.

1. What is $\frac{2}{3}$ of $\frac{3}{4}$ of $\frac{5}{6}$ of 48?

- (A) 30 (B) 20
(C) 24 (D) 16

2. A water tank is $\frac{5}{8}$ full. After $\frac{1}{3}$ of the water in it is used, what fraction of the FULL tank's capacity is now in the tank?

- (A) $\frac{5}{24}$ (B) $\frac{7}{24}$
(C) $\frac{5}{12}$ (D) $\frac{1}{3}$

3. By what number must $\frac{4}{9}$ be multiplied so the product is $\frac{8}{3}$?

- (A) $\frac{32}{27}$ (B) $\frac{3}{2}$
(C) $\frac{1}{6}$ (D) 6

4. Which of these is the LARGEST: $\frac{7}{8}$, $\frac{11}{12}$, $\frac{5}{6}$, or $\frac{9}{10}$?

- (A) $\frac{9}{10}$ (B) $\frac{11}{12}$
(C) $\frac{7}{8}$ (D) $\frac{5}{6}$

5. A recipe needs $2\frac{1}{4}$ cups of flour to make 3 loaves. How many cups are needed for 5 loaves?

(A) $11\frac{1}{4}$

(B) $\frac{3}{4}$

(C) $4\frac{1}{2}$

(D) $3\frac{3}{4}$

6. Ravi gives away $\frac{1}{2}$ of his marbles, then $\frac{1}{3}$ of what remains, then $\frac{1}{4}$ of what is left after that. What fraction of his original marbles does he still have?

(A) $\frac{1}{24}$

(B) $\frac{1}{12}$

(C) $\frac{5}{12}$

(D) $\frac{1}{4}$

7. Compute $(2\frac{1}{2} \times 1\frac{3}{5}) \div 1\frac{1}{3}$.

(A) $\frac{16}{3}$

(B) 3

(C) 4

(D) $\frac{27}{16}$

8. How many $\frac{2}{3}$ -litre bottles can be completely filled from $7\frac{1}{2}$ litres of milk?

(A) 5

(B) 12

(C) 11

(D) 10

9. Without exact computation, which statement about $24 \div \frac{5}{6}$ is TRUE?

(A) It is greater than 24.

(B) It equals 24.

(C) It is less than 24 but more than 20.

(D) It is less than $\frac{5}{6}$.

10. Order these from SMALLEST to LARGEST: $\frac{3}{5}$, $\frac{5}{8}$, $\frac{7}{10}$. Which is the smallest?

(A) $\frac{3}{5}$

(B) $\frac{5}{8}$

(C) $\frac{7}{10}$

(D) they are all equal

11. A car travels at $2\frac{2}{5}$ km per minute. How far does it go in $3\frac{1}{3}$ minutes?

(A) $6\frac{6}{15}$ km

(B) $5\frac{11}{15}$ km

(C) $\frac{12}{25}$ km

(D) 8 km

12. A field is $\frac{3}{4}$ of an acre. One-third of the field is for wheat and $\frac{2}{5}$ of the REMAINING field is for rice. What fraction of an acre is for rice?

(A) $\frac{3}{10}$

(B) $\frac{1}{5}$

(C) $\frac{1}{2}$

(D) $\frac{9}{20}$

13. Spending $\frac{2}{5}$ of her savings, Meena is left with Rs 360. What were her savings in rupees?

(A) 600

(B) 900

(C) 144

(D) 504

14. Compute $\frac{5}{6} \times \frac{3}{10} \div \frac{1}{4}$.

(A) $\frac{1}{16}$

(B) $\frac{5}{16}$

(C) $\frac{16}{25}$

(D) 1

15. Which is the correct ascending order of $\frac{5}{12}$, $\frac{1}{2}$, $\frac{3}{8}$?

(A) $\frac{5}{12} < \frac{3}{8} < \frac{1}{2}$

(B) $\frac{3}{8} < \frac{5}{12} < \frac{1}{2}$

(C) $\frac{1}{2} < \frac{5}{12} < \frac{3}{8}$

(D) $\frac{3}{8} < \frac{1}{2} < \frac{5}{12}$

- 16.** A rope is $4\frac{1}{2}$ m long. It is cut into pieces, each $\frac{3}{8}$ m long. How many full pieces are obtained, and what length (in m) is left over?
- (A) 12 pieces, $\frac{3}{8}$ m left (B) 11 pieces, $\frac{1}{8}$ m left
(C) 12 pieces, 0 m left (D) 16 pieces, 0 m left
- 17.** Which statement is TRUE?
- (A) Dividing 10 by $\frac{2}{5}$ gives a number larger than dividing 10 by $\frac{5}{2}$.
(B) Multiplying 10 by $\frac{2}{5}$ gives a number larger than 10.
(C) Dividing 10 by $\frac{5}{2}$ gives 25. (D) $10 \div \frac{2}{5}$ equals $10 \times \frac{2}{5}$.
- 18.** A tap fills $2\frac{1}{4}$ buckets in $\frac{3}{4}$ of an hour. How many buckets does it fill in 1 full hour?
- (A) $1\frac{11}{16}$ (B) 3
(C) $\frac{27}{16}$ (D) $\frac{9}{4}$
- 19.** Of a class, $\frac{3}{8}$ are boys. Of the boys, $\frac{2}{3}$ play football, and of the girls, $\frac{1}{4}$ play football. What fraction of the WHOLE class plays football?
- (A) $\frac{13}{32}$ (B) $\frac{11}{12}$
(C) $\frac{1}{4}$ (D) $\frac{5}{12}$
- 20.** After selling $\frac{3}{7}$ of his apples, then $\frac{1}{2}$ of the remainder, a vendor has 20 apples left. How many did he start with?
- (A) 140 (B) 35
(C) 70 (D) 60
- 21.** Compute $3\frac{1}{3} \times 2\frac{2}{5} \times \frac{1}{4}$.
- (A) 8 (B) $\frac{32}{15}$
(C) $6\frac{4}{5}$ (D) 2
- 22.** Three pipes can fill a tank in $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{6}$ of an hour each (working alone). Which fills the tank FASTEST?
- (A) the pipe taking $\frac{1}{2}$ hour (B) the pipe taking $\frac{1}{6}$ hour
(C) the pipe taking $\frac{1}{3}$ hour (D) they all fill it at the same time
- 23.** A wall is $\frac{4}{5}$ painted. If a painter does $\frac{1}{4}$ of the REMAINING wall, what fraction of the whole wall is still unpainted?
- (A) $\frac{1}{4}$ (B) $\frac{3}{20}$
(C) $\frac{1}{5}$ (D) $\frac{1}{20}$
- 24.** Half of a number added to one-third of the same number equals 25. The number is:
- (A) 15 (B) 60
(C) 30 (D) 50
- 25.** Compute $(\frac{4}{5} \div \frac{2}{3}) \times (\frac{5}{6} \div \frac{1}{4})$.
- (A) $\frac{1}{4}$ (B) 4
(C) $\frac{8}{45}$ (D) $\frac{10}{9}$

26. A book has 320 pages. Aman reads $\frac{3}{8}$ of it on the first day and $\frac{2}{5}$ of the remaining on the second day. How many pages are unread after two days?

- (A) 200 (B) 80
(C) 100 (D) 120

27. Which is closer to 1: $\frac{8}{9}$ or $\frac{9}{8}$?

- (A) $\frac{9}{8}$, because it is larger than 1. (B) They are equally close to 1.
(C) $\frac{8}{9}$, because 8 is less than 9.
(D) $\frac{8}{9}$, because its distance from 1 is $\frac{1}{9}$ while $\frac{9}{8}$'s distance is $\frac{1}{8}$.

28. By what number must $7\frac{1}{2}$ be divided to get $\frac{2}{3}$?

- (A) 5 (B) $\frac{5}{9}$
(C) $\frac{45}{4}$ (D) $\frac{15}{4}$

29. A garden's area is $\frac{7}{8}$ of a square plot of side $\frac{4}{5}$ m. The garden area (in m^2) is:

- (A) $\frac{14}{25}$ (B) $\frac{7}{10}$
(C) $\frac{16}{25}$ (D) $\frac{7}{8}$

30. Without computing the exact value, which is true about $\frac{18}{19} \times \frac{20}{21}$?

- (A) It is greater than 1. (B) It is greater than $\frac{20}{21}$.
(C) It is less than $\frac{18}{19}$. (D) It equals $\frac{18}{19}$.

31. Two-fifths of a pole is below ground and 5 m is above ground. How long is the pole (in m)?

- (A) 2 (B) $12\frac{1}{2}$
(C) $\frac{25}{3}$ (D) $\frac{10}{3}$

32. Compute $\frac{2}{3} \div (\frac{4}{5} \times \frac{5}{8})$.

- (A) $\frac{1}{3}$ (B) $\frac{16}{15}$
(C) $\frac{5}{12}$ (D) $\frac{4}{3}$

33. A jar of sweets is shared so that the elder child gets $\frac{3}{5}$ and the younger child gets $\frac{2}{3}$ of what is LEFT. If the younger child gets 8 sweets, how many sweets were in the jar?

- (A) 20 (B) 30
(C) 40 (D) 12

34. Which expression has the LARGEST value: $10 \times \frac{4}{5}$, $10 \div \frac{2}{3}$, $10 \times \frac{5}{4}$, or $10 \div \frac{5}{4}$?

- (A) $10 \div \frac{2}{3}$ (B) $10 \times \frac{5}{4}$
(C) $10 \times \frac{4}{5}$ (D) $10 \div \frac{5}{4}$

35. A recipe uses $\frac{3}{4}$ cup sugar for every $\frac{2}{3}$ cup flour. How many cups of sugar are needed for 2 cups of flour?

- (A) $\frac{1}{2}$ (B) $\frac{8}{9}$
(C) $\frac{3}{2}$ (D) $\frac{9}{4}$

36. Half of one-third of a quantity equals one-fourth of another quantity. If the first quantity is 24, the second quantity is:

- (A) 16 (B) 12
(C) 8 (D) 32

37. A $2\frac{1}{2}$ kg bag of rice is repacked into $\frac{5}{8}$ kg packets. How many full packets are made?

- (A) 3 (B) 4
(C) 5 (D) $\frac{25}{16}$

38. On a number line, which fraction lies exactly halfway between $\frac{1}{3}$ and $\frac{1}{2}$?

- (A) $\frac{2}{5}$ (B) $\frac{5}{12}$
(C) $\frac{1}{4}$ (D) $\frac{5}{6}$

39. A worker completes $\frac{1}{6}$ of a job each day. After working 2 days, he speeds up to $\frac{1}{4}$ of the remaining job per day. What fraction of the whole job is done by the end of the third day?

- (A) $\frac{7}{12}$ (B) $\frac{5}{12}$
(C) $\frac{5}{8}$ (D) $\frac{1}{2}$

40. Compute $1\frac{1}{5} \div \frac{3}{10} \div 2$.

- (A) 8 (B) $\frac{1}{2}$
(C) 2 (D) $\frac{9}{25}$

41. A tank loses $\frac{1}{8}$ of its water each hour through evaporation. If it starts full, what fraction of the original water remains after 2 hours?

- (A) $\frac{49}{64}$ (B) $\frac{3}{4}$
(C) $\frac{1}{16}$ (D) $\frac{7}{8}$

42. If $\frac{5}{6}$ of a sum of money is Rs 250, then $\frac{2}{3}$ of the same sum is:

- (A) Rs 300 (B) Rs $166\frac{2}{3}$
(C) Rs 200 (D) Rs 150

43. Compute $(\frac{3}{4} - \frac{1}{6}) \div \frac{7}{12}$.

- (A) $\frac{7}{24}$ (B) $\frac{49}{144}$
(C) 1 (D) $\frac{12}{7}$

44. A pizza is cut so that Anya takes $\frac{1}{3}$, Bina takes $\frac{1}{4}$ of the WHOLE pizza, and Chetan takes $\frac{1}{2}$ of WHAT REMAINS. What fraction of the pizza is left after all three?

- (A) $\frac{5}{12}$ (B) $\frac{5}{24}$
(C) $\frac{1}{4}$ (D) $\frac{1}{6}$

45. Which of these equals 6: $\frac{1}{3} \div \frac{1}{18}$, $\frac{1}{3} \times 6$, $18 \div \frac{1}{3}$, or $\frac{1}{18} \div \frac{1}{3}$?

- (A) $\frac{1}{3} \div \frac{1}{18}$ (B) $18 \div \frac{1}{3}$
(C) $\frac{1}{3} \times 6$ (D) $\frac{1}{18} \div \frac{1}{3}$

46. A square has area $9/16 \text{ m}^2$. What is the length of its side (in m)?

- (A) $3/4$ (B) $9/32$
(C) $9/8$ (D) $81/256$

47. Tap A fills $3/5$ of a tank in 1 hour and tap B fills $1/4$ of the tank in 1 hour. Working together for 1 hour, what fraction of the tank is filled?

- (A) $3/20$ (B) $4/9$
(C) $17/20$ (D) 1

48. A merchant has $12 \frac{1}{2} \text{ m}$ of cloth. He sells $2/5$ of it, then $1/3$ of the remainder. How many metres are left?

- (A) $7 \frac{1}{2}$ (B) 5
(C) $2 \frac{1}{2}$ (D) $4 \frac{1}{6}$

49. Half of three-quarters of a number is 9. The number is:

- (A) 12 (B) 48
(C) 24 (D) $27/2$

50. Compute $4 \div 2/3 \div 3/2$.

- (A) 4 (B) 1
(C) 9 (D) $16/9$

51. A school has 600 students. $5/12$ are in the junior section. Of the juniors, $2/5$ are girls. How many junior girls are there?

- (A) 250 (B) 240
(C) 144 (D) 100

52. Which is larger and by how much: $3/4$ of $5/6$, or $5/6$ of $3/4$?

- (A) They are equal. (B) $3/4$ of $5/6$ is larger.
(C) $5/6$ of $3/4$ is larger. (D) They differ by $1/12$.

53. A $3 \frac{3}{4}$ litre container of paint covers $5/8$ of a room with one coat. How many litres are needed to cover the WHOLE room with one coat?

- (A) 6 (B) $2 \frac{11}{32}$
(C) $75/32$ (D) 5

54. Which statement is FALSE?

- (A) Dividing a whole number by a fraction less than 1 gives a larger number.
(B) Multiplying by a fraction less than 1 gives a smaller number.
(C) Dividing a whole number by a fraction greater than 1 always gives a larger number.
(D) The reciprocal of a proper fraction is greater than 1.

55. A vessel is $2/3$ full of water. After 6 litres are added, it is $5/6$ full. What is the vessel's full capacity (in litres)?

(A) 12

(B) 36

(C) 24

(D) 30

56. Compute $2\frac{2}{3} \div 1\frac{1}{9}$ and write it as a mixed number.

(A) $2\frac{22}{27}$

(B) $\frac{5}{12}$

(C) $2\frac{1}{6}$

(D) $2\frac{2}{5}$

57. A field is divided into three parts. The first is $\frac{2}{5}$ of the field, the second is $\frac{1}{3}$ of the field, and the third is the rest. The third part is what fraction of the field?

(A) $\frac{1}{15}$

(B) $\frac{11}{15}$

(C) $\frac{4}{15}$

(D) $\frac{1}{8}$

58. A photograph $\frac{4}{5}$ m wide is enlarged to 3 times its width. After enlargement, by what fraction of a metre is the width MORE than 2 m?

(A) $\frac{12}{5}$

(B) $\frac{2}{15}$

(C) $\frac{1}{5}$

(D) $\frac{2}{5}$

59. Compute $\frac{1}{2}$ of $4 \div \frac{1}{3}$ of 6.

(A) 4

(B) $\frac{1}{9}$

(C) 9

(D) 1

60. In a basket, $\frac{5}{9}$ of the fruits are apples and the rest are oranges. If the apples number 40, how many oranges are there?

(A) 32

(B) 72

(C) 45

(D) 40

Solutions — Mathematics (Class 7), Chapter 8:

Working with Fractions — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	D	21	D	31	C	41	A	51	D
2	C	12	B	22	B	32	D	42	C	52	A
3	D	13	A	23	B	33	B	43	C	53	A
4	B	14	D	24	C	34	A	44	B	54	C
5	D	15	B	25	B	35	D	45	A	55	B
6	D	16	C	26	D	36	A	46	A	56	D
7	B	17	A	27	D	37	B	47	C	57	C
8	C	18	B	28	C	38	B	48	B	58	D
9	A	19	A	29	A	39	D	49	C	59	D
10	A	20	C	30	C	40	C	50	A	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Work left to right, or multiply all the fractions first: $\frac{2}{3} \times \frac{3}{4} \times \frac{5}{6} = \frac{(2 \times 3 \times 5)}{(3 \times 4 \times 6)} = \frac{30}{72} = \frac{5}{12}$. Then $\frac{5}{12}$ of $48 = 48 \div 12 \times 5 = 4 \times 5 = 20$. A reader who computes only $\frac{5}{6}$ of $48 = 40$ then halves carelessly might land on 24 or 16; multiplying the wrong combination of numerators gives 30. The chained 'of' simply means multiply everything, so 20 is correct.

2. (C) If $\frac{1}{3}$ of the water is used, then $\frac{2}{3}$ of the water remains. The water present is $\frac{2}{3}$ of $\frac{5}{8} = \frac{(2 \times 5)}{(3 \times 8)} = \frac{10}{24} = \frac{5}{12}$ of the full capacity. Subtracting $\frac{1}{3}$ from $\frac{5}{8}$ directly (a common slip) wrongly treats the $\frac{1}{3}$ as a fraction of the whole tank rather than of the water; multiplying $\frac{1}{3} \times \frac{5}{8}$ gives the amount removed ($\frac{5}{24}$), not what remains. So $\frac{5}{12}$ stays.

3. (D) We need x with $\frac{4}{9} \times x = \frac{8}{3}$, so $x = (\frac{8}{3}) \div (\frac{4}{9}) = \frac{8}{3} \times \frac{9}{4} = \frac{72}{12} = 6$. Multiplying instead of dividing gives $\frac{8}{3} \times \frac{4}{9} = \frac{32}{27}$; flipping the wrong fraction gives

$8/3 \times 9/4$... only the correct flip of the divisor $4/9$ to $9/4$ yields 6. So the required multiplier is 6.

4. (B) Each fraction is just below 1; the largest is the one whose 'missing piece' from 1 is smallest. The gaps are $1/8$, $1/12$, $1/6$ and $1/10$. The smallest gap is $1/12$, so $11/12$ is closest to 1 and therefore largest. Picking $9/10$ confuses a small denominator-numerator gap; $1/10$ is bigger than $1/12$, so $9/10$ is not the largest. Hence $11/12$.

5. (D) Per loaf the flour is $2 \frac{1}{4} \div 3 = 9/4 \div 3 = 9/12 = 3/4$ cup. For 5 loaves: $3/4 \times 5 = 15/4 = 3 \frac{3}{4}$ cups. Multiplying $2 \frac{1}{4}$ by 5 without dividing by 3 gives $11 \frac{1}{4}$; stopping at the per-loaf amount gives $3/4$. The two-step rate reasoning gives $3 \frac{3}{4}$ cups.

6. (D) Each step keeps the complement: after the first, $1/2$ remains; after the second, $2/3$ of that = $1/3$; after the third, $3/4$ of that = $1/4$. So $1/2 \times 2/3 \times 3/4 = 6/24 = 1/4$ remains. Adding the fractions given away ($1/2 + 1/3 + 1/4$) is wrong because each acts on a smaller leftover. Multiplying the given-away fractions ($1/2 \times 1/3 \times 1/4 = 1/24$) is also wrong. The fraction kept is $1/4$.

7. (B) Convert: $2 \frac{1}{2} = 5/2$, $1 \frac{3}{5} = 8/5$, $1 \frac{1}{3} = 4/3$. First $5/2 \times 8/5 = 40/10 = 4$. Then $4 \div 4/3 = 4 \times 3/4 = 3$. Multiplying by $4/3$ instead of dividing gives $16/3$; forgetting to convert mixed numbers throws off the rest. The value is 3.

8. (C) Divide total by bottle size: $7 \frac{1}{2} \div 2/3 = 15/2 \times 3/2 = 45/4 = 11 \frac{1}{4}$. Only complete bottles count, so 11 bottles can be filled (with $1/4$ of a bottle's worth left over). Multiplying $15/2 \times 2/3 = 5$ comes from forgetting to invert; rounding $11 \frac{1}{4}$ up to 12 over-counts since the last bottle is not full. So 11 complete bottles.

9. (A) Dividing by a fraction less than 1 always makes a positive number bigger, because you are asking how many $5/6$ -sized pieces fit in 24, and each piece is smaller than 1. Indeed $24 \div 5/6 = 24 \times 6/5 = 144/5 = 28.8$, which exceeds 24. The trap is treating division like multiplication (which would shrink it). So the result is greater than 24.

10. (A) Use a common denominator of 40: $3/5 = 24/40$, $5/8 = 25/40$, $7/10 = 28/40$. The smallest is $24/40 = 3/5$. Judging by numerator alone (7 is biggest) or denominator alone misleads. So $3/5$ is smallest.

11. (D) Convert: $2 \frac{2}{5} = 12/5$, $3 \frac{1}{3} = 10/3$. Distance = $12/5 \times 10/3 = 120/15 = 8$ km. Adding the mixed numbers ($\approx 5 \frac{11}{15}$) confuses 'per minute $\times$ minutes' with addition; multiplying only the fraction parts gives $12/25$. The distance is 8 km.

12. (B) Wheat uses $1/3$ of $3/4 = 1/4$ acre, leaving $3/4 - 1/4 = 1/2$ acre. Rice is $2/5$ of that remaining $1/2$ acre = $2/5 \times 1/2 = 1/5$ acre. Taking $2/5$ of the whole $3/4$ (= $3/10$) ignores 'of the remaining'; taking $2/5$ of the original field after wheat without recomputing the leftover slips too. Rice covers $1/5$ acre.

13. (A) After spending $\frac{2}{5}$, the fraction left is $\frac{3}{5}$. So $\frac{3}{5}$ of savings = 360, giving savings = $360 \div \frac{3}{5} = 360 \times \frac{5}{3} = 600$. Treating 360 as the $\frac{2}{5}$ part gives $360 \times \frac{5}{2} = 900$; taking $\frac{2}{5}$ of 360 gives 144. The savings are Rs 600.

14. (D) Do multiplication and division left to right. $\frac{5}{6} \times \frac{3}{10} = \frac{15}{60} = \frac{1}{4}$. Then $\frac{1}{4} \div \frac{1}{4} = \frac{1}{4} \times 4 = 1$. Multiplying by $\frac{1}{4}$ instead of dividing gives $\frac{1}{16}$; mishandling the order also misleads. The value is 1.

15. (B) Common denominator 24: $\frac{5}{12} = \frac{10}{24}$, $\frac{1}{2} = \frac{12}{24}$, $\frac{3}{8} = \frac{9}{24}$. Ascending: $\frac{9}{24}$, $\frac{10}{24}$, $\frac{12}{24}$, i.e. $\frac{3}{8} < \frac{5}{12} < \frac{1}{2}$. Comparing numerators or denominators alone scrambles the order. The correct order is $\frac{3}{8} < \frac{5}{12} < \frac{1}{2}$.

16. (C) $4 \frac{1}{2} \div \frac{3}{8} = \frac{9}{2} \times \frac{8}{3} = \frac{72}{6} = 12$ exactly. So 12 full pieces with nothing left over. Multiplying $\frac{9}{2} \times \frac{3}{8} = \frac{27}{16}$ (forgetting to invert) gives a meaningless count; the exact quotient 12 means no remainder. Answer: 12 pieces, 0 m left.

17. (A) $10 \div \frac{2}{5} = 10 \times \frac{5}{2} = 25$, while $10 \div \frac{5}{2} = 10 \times \frac{2}{5} = 4$; $25 > 4$, so dividing by the smaller fraction gives the larger result — true. Multiplying 10 by $\frac{2}{5}$ (a fraction below 1) shrinks it to 4, not larger. $10 \div \frac{5}{2} = 4$, not 25. And dividing is never the same as multiplying by the same fraction. The first statement is correct.

18. (B) Rate = buckets $\div$ time = $2 \frac{1}{4} \div \frac{3}{4} = \frac{9}{4} \times \frac{4}{3} = \frac{36}{12} = 3$ buckets per hour. Multiplying $\frac{9}{4} \times \frac{3}{4} = \frac{27}{16}$ (no inversion) is wrong; $2 \frac{1}{4}$ itself is the amount in only $\frac{3}{4}$ hour, not a full hour. The rate is 3 buckets per hour.

19. (A) Express both groups as a fraction of the class out of 32. Boys = $\frac{3}{8} = \frac{12}{32}$, and footballers among boys = $\frac{2}{3} \times \frac{12}{32} = \frac{8}{32}$. Girls = $\frac{5}{8} = \frac{20}{32}$, and footballers among girls = $\frac{1}{4} \times \frac{20}{32} = \frac{5}{32}$. Total footballers = $\frac{8}{32} + \frac{5}{32} = \frac{13}{32}$ of the class. Adding $\frac{2}{3} + \frac{1}{4}$ directly (= $\frac{11}{12}$) ignores that each acts on a different group; counting only the boys' footballers gives $\frac{1}{4}$. The fraction is $\frac{13}{32}$.

20. (C) After selling $\frac{3}{7}$, then $\frac{4}{7}$ remain; selling half of that remainder leaves $\frac{1}{2} \times \frac{4}{7} = \frac{2}{7}$ of the original. So $\frac{2}{7}$ of the start = 20, giving start = $20 \div \frac{2}{7} = 20 \times \frac{7}{2} = 70$. Treating 20 as $\frac{4}{7}$ (the first remainder) gives 35; doubling once more gives 140. He started with 70 apples.

21. (D) Convert: $3 \frac{1}{3} = \frac{10}{3}$, $2 \frac{2}{5} = \frac{12}{5}$. Then $\frac{10}{3} \times \frac{12}{5} = \frac{120}{15} = 8$, and $8 \times \frac{1}{4} = 2$. Forgetting the final $\times \frac{1}{4}$ leaves 8; multiplying only fraction parts ($\frac{1}{3} \times \frac{2}{5} \times \frac{1}{4}$) gives $\frac{1}{30}$ which is far off. The product is 2.

22. (B) Fastest means the smallest time. Comparing $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{6}$: the smallest is $\frac{1}{6}$ (since $\frac{1}{6} < \frac{1}{3} < \frac{1}{2}$). A common trap is to think the bigger-looking fraction $\frac{1}{2}$ is fastest, but more time means slower. The $\frac{1}{6}$ -hour pipe is fastest.

- 23. (B)** Remaining before the painter starts = $1 - 4/5 = 1/5$. He paints $1/4$ of that $1/5 = 1/20$, leaving $1/5 - 1/20 = 4/20 - 1/20 = 3/20$ unpainted. Taking $1/4$ of the whole wall, or forgetting to subtract from the $1/5$ leftover, gives $1/4$ or $1/5$. Unpainted = $3/20$.
- 24. (C)** Half plus a third of the number is $(1/2 + 1/3) = 5/6$ of the number. So $5/6$ of $n = 25$, giving $n = 25 \div 5/6 = 25 \times 6/5 = 30$. Adding $1/2 + 1/3$ as $2/5$ (denominator slip) misleads; treating 25 as the answer directly is wrong. The number is 30.
- 25. (B)** First bracket: $4/5 \div 2/3 = 4/5 \times 3/2 = 12/10 = 6/5$. Second bracket: $5/6 \div 1/4 = 5/6 \times 4 = 20/6 = 10/3$. Product = $6/5 \times 10/3 = 60/15 = 4$. Multiplying without inverting either divisor gives small wrong values like $8/45$. The answer is 4.
- 26. (D)** Day 1 reads $3/8$ of $320 = 120$, leaving 200. Day 2 reads $2/5$ of $200 = 80$, leaving $200 - 80 = 120$ unread. Taking $2/5$ of the whole 320 (=128) ignores 'of the remaining'; stopping at 200 forgets day 2. Unread pages = 120.
- 27. (D)** Distance from 1 is $|1 - 8/9| = 1/9$ and $|9/8 - 1| = 1/8$. Since $1/9 < 1/8$, the fraction $8/9$ is closer to 1. Being larger than 1 (as $9/8$ is) does not make it closer. So $8/9$ is closer, by exactly $1/9$ versus $1/8$.
- 28. (C)** We need x with $15/2 \div x = 2/3$, so $x = (15/2) \div (2/3) = 15/2 \times 3/2 = 45/4 = 11 \frac{1}{4}$. A reader who computes $15/2 \times 2/3 = 5$ has inverted the wrong fraction; the dividend stays put, the divisor side is rearranged. So $x = 45/4$.
- 29. (A)** Square's area = $(4/5)^2 = 16/25 \text{ m}^2$. Garden = $7/8$ of $16/25 = (7 \times 16)/(8 \times 25) = 112/200 = 14/25 \text{ m}^2$. Forgetting to square the side (using $4/5$ directly) gives $7/10$; stopping at $16/25$ forgets the $7/8$ factor. The garden area is $14/25 \text{ m}^2$.
- 30. (C)** Multiplying $18/19$ by $20/21$ — a factor less than 1 — must shrink it below $18/19$. Likewise the product is below $20/21$. Since both factors are under 1, the product is under 1, not greater. So it is less than $18/19$.
- 31. (C)** Below ground = $2/5$, so above ground = $3/5$ of the pole = 5 m. Pole = $5 \div 3/5 = 5 \times 5/3 = 25/3 = 8 \frac{1}{3} \text{ m}$. Treating 5 m as the $2/5$ part gives $5 \times 5/2 = 12 \frac{1}{2}$; taking $2/5$ of 5 gives 2. The pole is $25/3 \text{ m}$ long.
- 32. (D)** First the bracket: $4/5 \times 5/8 = 20/40 = 1/2$. Then $2/3 \div 1/2 = 2/3 \times 2 = 4/3$. Dividing before doing the bracket, or multiplying by $1/2$ instead of dividing, gives $1/3$. The value is $4/3$.
- 33. (B)** After the elder takes $3/5$, the remainder is $2/5$ of the jar. The younger takes $2/3$ of that = $2/3 \times 2/5 = 4/15$ of the jar = 8 sweets. So jar = $8 \div 4/15 = 8 \times 15/4 = 30$. Treating 8 as $2/5$ of the jar (=20) skips the elder's share interaction. The jar had 30 sweets.

34. (A) Compute each: $10 \times 4/5 = 8$; $10 \div 2/3 = 10 \times 3/2 = 15$; $10 \times 5/4 = 12.5$; $10 \div 5/4 = 10 \times 4/5 = 8$. The largest is 15, from $10 \div 2/3$ — dividing by the smallest fraction enlarges the most. Multiplying by $5/4$ only reaches 12.5, and the others are 8. The largest value is 15 from $10 \div 2/3$.

35. (D) Sugar per cup of flour = $3/4 \div 2/3 = 3/4 \times 3/2 = 9/8$ cup. For 2 cups flour: $9/8 \times 2 = 18/8 = 9/4 = 2 \frac{1}{4}$ cups. Multiplying $3/4 \times 2/3 (=1/2)$ confuses the ratio direction. Sugar needed = $9/4$ cups.

36. (A) Half of a third of 24 = $1/2 \times 1/3 \times 24 = 1/6 \times 24 = 4$. This equals $1/4$ of the second, so $1/4$ of second = 4, giving second = $4 \times 4 = 16$. Computing $1/6$ of 24 wrong, or dividing instead of multiplying at the end, gives 12 or 8. The second quantity is 16.

37. (B) $2 \frac{1}{2} \div 5/8 = 5/2 \times 8/5 = 40/10 = 4$ exactly. So 4 full packets with nothing left. Multiplying $5/2 \times 5/8 = 25/16$ (no inversion) is wrong; the clean quotient 4 means no leftover. Answer: 4 packets.

38. (B) The midpoint is the average: $(1/3 + 1/2) \div 2 = (2/6 + 3/6) \div 2 = (5/6) \div 2 = 5/12$. Averaging numerators and denominators separately gives $2/5$, a classic error that is not the true midpoint. The midpoint is $5/12$.

39. (D) In 2 days he does $2 \times 1/6 = 1/3$, leaving $2/3$. On day 3 he does $1/4$ of the remaining $2/3 = 1/4 \times 2/3 = 1/6$. Total done = $1/3 + 1/6 = 2/6 + 1/6 = 3/6 = 1/2$. Taking $1/4$ of the whole job on day 3 ($=1/4$) instead of of the remainder gives $7/12$. By day 3's end, $1/2$ is done.

40. (C) Left to right: $1 \frac{1}{5} = 6/5$; $6/5 \div 3/10 = 6/5 \times 10/3 = 60/15 = 4$; then $4 \div 2 = 2$. Multiplying by 2 at the end (instead of dividing) gives 8; mishandling the first division gives $9/25$. The value is 2.

41. (A) Each hour $7/8$ of the current water remains. After 2 hours: $7/8 \times 7/8 = 49/64$ of the original. Subtracting $1/8$ twice (giving $3/4$) is wrong because the second hour's loss is $1/8$ of a smaller amount, not of the original. Remaining = $49/64$.

42. (C) First find the whole: sum = $250 \div 5/6 = 250 \times 6/5 = 300$. Then $2/3$ of 300 = 200. Taking $2/3$ of 250 directly (≈ 166.67) ignores that 250 is only $5/6$ of the sum. The answer is Rs 200.

43. (C) Bracket first: $3/4 - 1/6 = 9/12 - 2/12 = 7/12$. Then $7/12 \div 7/12 = 1$. Multiplying $7/12 \times 7/12$ (no inversion) gives $49/144$; subtracting incorrectly throws off the bracket. The value is 1.

44. (B) Anya + Bina take $1/3 + 1/4 = 4/12 + 3/12 = 7/12$, leaving $5/12$. Chetan takes $1/2$ of that $5/12 = 5/24$, leaving $5/12 - 5/24 = 10/24 - 5/24 = 5/24$. Taking $1/2$ of the whole pizza for Chetan over-counts. Left over = $5/24$.

45. (A) Evaluate each: $1/3 \div 1/18 = 1/3 \times 18 = 6$ ✓. Meanwhile $1/3 \times 6 = 2$, $18 \div 1/3 = 18 \times 3 = 54$, and $1/18 \div 1/3 = 1/18 \times 3 = 1/6$. Only the first, which uses invert-and-multiply on the divisor $1/18$, gives 6. The others give 2, 54 and $1/6$.

46. (A) The side is the number whose square is $9/16$. Since $(3/4)^2 = 9/16$, the side is $3/4$ m. Halving the area ($9/32$) or squaring again ($81/256$) are wrong operations; taking a side means the square root of the area. Side = $3/4$ m.

47. (C) Combined rate adds: $3/5 + 1/4 = 12/20 + 5/20 = 17/20$ of the tank per hour. So in 1 hour they fill $17/20$. Multiplying the rates ($3/20$) instead of adding is the trap; the total is under 1, so it is not full. Filled = $17/20$.

48. (B) Sells $2/5$ of $12 \frac{1}{2} = 2/5 \times 25/2 = 5$ m, leaving $7 \frac{1}{2}$ m. Then sells $1/3$ of $7 \frac{1}{2} = 2 \frac{1}{2}$ m, leaving $7 \frac{1}{2} - 2 \frac{1}{2} = 5$ m. Stopping after the first sale leaves $7 \frac{1}{2}$ (a partial answer); taking $1/3$ of the whole skips the remainder logic. Left = 5 m.

49. (C) Half of three-quarters is $1/2 \times 3/4 = 3/8$ of the number. So $3/8$ of $n = 9$, giving $n = 9 \div 3/8 = 9 \times 8/3 = 24$. Computing $3/8$ of 9 (≈ 3.375) or dividing in the wrong direction gives $27/2$ or 12. The number is 24.

50. (A) Left to right: $4 \div 2/3 = 4 \times 3/2 = 6$; then $6 \div 3/2 = 6 \times 2/3 = 4$. Multiplying instead of inverting at either step distorts it; doing $4 \div (2/3 \div 3/2)$ changes grouping. As written, the value is 4.

51. (D) Juniors = $5/12$ of 600 = 250. Junior girls = $2/5$ of 250 = 100. Taking $2/5$ of the whole 600 (=240) ignores the junior restriction; stopping at 250 forgets the girls fraction. Junior girls = 100.

52. (A) Both equal $3/4 \times 5/6 = 15/24 = 5/8$, because multiplication is commutative: 'a of b' equals 'b of a'. So neither is larger; they are equal at $5/8$. The order of 'of' does not change a product.

53. (A) If $3 \frac{3}{4} = 15/4$ litres covers $5/8$ of the room, then $1/8$ covers $(15/4) \div 5 = 3/4$ litre, and the whole ($8/8$) needs $3/4 \times 8 = 6$ litres. Equivalently $15/4 \div 5/8 = 15/4 \times 8/5 = 6$. Multiplying $15/4 \times 5/8 = 75/32$ (no inversion) is wrong. Needed = 6 litres.

54. (C) Dividing by a fraction GREATER than 1 actually SHRINKS the number (e.g. $6 \div 3/2 = 4 < 6$), so that statement is false. The others are true: dividing by a fraction below 1 enlarges ($6 \div 1/2 = 12$), multiplying by a fraction below 1 shrinks, and a proper fraction's reciprocal exceeds 1. The false claim is the one about dividing by a fraction greater than 1.

55. (B) Adding 6 litres raised the level by $5/6 - 2/3 = 5/6 - 4/6 = 1/6$ of capacity. So $1/6$ of capacity = 6, giving capacity = $6 \times 6 = 36$ litres. Treating 6 litres as the whole difference between full and empty, or as $1/3$, gives 12 or 24. Capacity = 36 litres.

56. (D) Convert: $2\frac{2}{3} = \frac{8}{3}$, $1\frac{1}{9} = \frac{10}{9}$. Then $\frac{8}{3} \div \frac{10}{9} = \frac{8}{3} \times \frac{9}{10} = \frac{72}{30} = \frac{12}{5} = 2\frac{2}{5}$. Multiplying $\frac{8}{3} \times \frac{10}{9} = \frac{80}{27}$ (no inversion) gives $2\frac{26}{27}$; dividing the wrong way gives $\frac{5}{12}$. The result is $2\frac{2}{5}$.

57. (C) First two parts: $\frac{2}{5} + \frac{1}{3} = \frac{6}{15} + \frac{5}{15} = \frac{11}{15}$. The third is $1 - \frac{11}{15} = \frac{4}{15}$. Adding $\frac{2}{5} + \frac{1}{3}$ as $\frac{3}{8}$ (denominator slip) gives $\frac{5}{8}$ leftover; the correct common denominator gives $\frac{4}{15}$. The third part is $\frac{4}{15}$.

58. (D) Enlarged width = $3 \times \frac{4}{5} = \frac{12}{5} = 2\frac{2}{5}$ m. The amount more than 2 m is $\frac{12}{5} - 2 = \frac{12}{5} - \frac{10}{5} = \frac{2}{5}$ m. Reporting the full enlarged width ($\frac{12}{5}$) skips the comparison to 2 m; multiplying widths gives nonsense. The excess over 2 m is $\frac{2}{5}$ m.

59. (D) Read each 'of' as multiply, then divide: ' $\frac{1}{2}$ of 4' = 2, and ' $\frac{1}{3}$ of 6' = 2, so $2 \div 2 = 1$. A reader who divides 4 by $\frac{1}{3}$ first (=12) then mishandles the rest gets 9 or other values. Computing each 'of' group before dividing gives 1.

60. (A) Apples are $\frac{5}{9}$ of the total, so total = $40 \div \frac{5}{9} = 40 \times \frac{9}{5} = 72$. Oranges are the rest = $72 - 40 = 32$ (which is $\frac{4}{9}$ of 72). Treating 40 as the whole gives 72; computing $\frac{5}{9}$ of 40 ignores that 40 IS the apple count. There are 32 oranges.